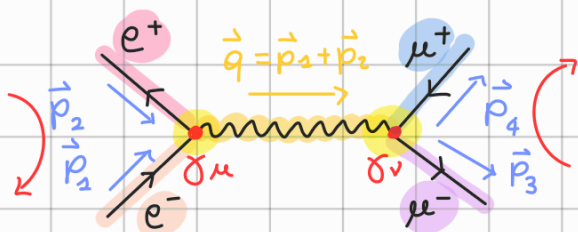


$$e^+ e^- \rightarrow \mu^+ \mu^-$$

1) we can draw this (this is a tool on internet)



$$q^2 = (p_1 + p_2)^2 = s$$

columns:

MATRIX ELEMENT (from Feynman rules)

$$M = \bar{v}(p_2, s_2) i e \gamma^\mu u(p_1, s_1) \left(-\frac{i \eta_{\mu\nu}}{q^2} \right) \bar{u}(p_3, s_3) i e \gamma^\nu v(p_4, s_4)$$

$$= \frac{i e^2}{s} \left[\underbrace{\bar{v}(p_2, s_2) \gamma^\mu u(p_1, s_1)}_{\text{electronic current}} \eta_{\mu\nu} \underbrace{\bar{u}(p_3, s_3) \gamma^\nu v(p_4, s_4)}_{\text{muonic current}} \right]$$

complex conj.

$$M^* = -\frac{i e^2}{s} \left[\bar{u}(p_1, s_1) \gamma^\rho v(p_2, s_2) \eta_{\rho\sigma} \bar{v}(p_4, s_4) \gamma^\sigma u(p_3, s_3) \right]$$

$$M^2 = M M^* = \frac{e^4}{s^2} \left[\bar{v}(p_2, s_2) \gamma^\mu u(p_1, s_1) \eta_{\mu\nu} \bar{u}(p_3, s_3) \gamma^\nu v(p_4, s_4) \right] \times$$

$$\left[\bar{u}(p_1, s_1) \gamma^\rho v(p_2, s_2) \eta_{\rho\sigma} \bar{v}(p_4, s_4) \gamma^\sigma u(p_3, s_3) \right]$$

calculate the average modulus squared

$$\rightarrow \langle |M|^2 \rangle = \frac{1}{4} \sum_{\text{spins}} \frac{e^4}{s^2} \left[\bar{v}(p_2, s_2) \gamma^\mu u(p_1, s_1) \eta_{\mu\nu} \bar{u}(p_3, s_3) \gamma^\nu v(p_4, s_4) \right] \times$$

$$\left[\bar{u}(p_1, s_1) \gamma^\rho v(p_2, s_2) \eta_{\rho\sigma} \bar{v}(p_4, s_4) \gamma^\sigma u(p_3, s_3) \right]$$

using

completeness relations

- $\sum_s u(p, s) \bar{u}(p, s) = \not{p} + m$
- $\sum_s v(p, s) \bar{v}(p, s) = \not{p} - m$

$$\not{p} = p_\mu \gamma^\mu$$

and trace properties
and considering
the approx
 $E \gg m_e \rightarrow m_e \approx 0$

$$= \frac{1}{4} \sum_{\text{spiu}} \frac{e^4}{s^2} \left[\bar{v}(p_2, s_2) \gamma^\mu u(p_1, s_1) \eta_{\mu\nu} \bar{u}(p_3, s_3) \gamma^\nu v(p_4, s_4) \right] \times$$

$$\left[\bar{u}(p_1, s_1) \gamma^\rho v(p_2, s_2) \eta_{\rho\sigma} \bar{v}(p_4, s_4) \gamma^\sigma u(p_3, s_3) \right]$$

$$= \frac{1}{4} \frac{e^4}{s^2} \sum_{\text{spiu}} \left[\bar{v}_2 \gamma^\mu u_1 \bar{u}_3 \gamma^\nu v_4 \bar{u}_1 \gamma^\rho v_2 \bar{v}_4 \gamma^\sigma u_3 \right] \eta_{\mu\nu} \eta_{\rho\sigma}$$

$$= \frac{1}{4} \frac{e^4}{s^2} \sum_{\text{spiu}} \left[\bar{u}_1 \gamma^\rho v_2 \bar{v}_2 \gamma^\mu u_1 \bar{u}_3 \gamma^\nu v_4 \bar{v}_4 \gamma^\sigma u_3 \right] \eta_{\mu\nu} \eta_{\rho\sigma}$$

$$= \frac{1}{4} \frac{e^4}{s^2} \sum_{s_1, s_3} \left[\bar{u}_1 \gamma^\rho (\not{p}_2 - m_e) \gamma^\mu u_1 \bar{u}_3 \gamma^\nu (\not{p}_4 - m_\mu) \gamma^\sigma u_3 \right] \eta_{\mu\nu} \eta_{\rho\sigma}$$

$$= \frac{1}{4} \frac{e^4}{s^2} \left[\underbrace{(\not{p}_1 + m_e) \gamma^\rho (\not{p}_2 - m_e) \gamma^\mu}_{\text{Tr}[(\not{p}_1 + m_e) \gamma^\rho (\not{p}_2 - m_e) \gamma^\mu]} \underbrace{(\not{p}_3 + m_\mu) \gamma^\nu (\not{p}_4 - m_\mu) \gamma^\sigma}_{\text{Tr}[(\not{p}_3 + m_\mu) \gamma^\nu (\not{p}_4 - m_\mu) \gamma^\sigma]} \right] \eta_{\mu\nu} \eta_{\rho\sigma}$$

APPROX.

$$= \frac{1}{4} \frac{e^4}{s^2} \text{Tr}[(\not{p}_1 + m_e) \gamma^\rho (\not{p}_2 - m_e) \gamma^\mu] \text{Tr}[(\not{p}_3 + m_\mu) \gamma^\nu (\not{p}_4 - m_\mu) \gamma^\sigma] \eta_{\mu\nu} \eta_{\rho\sigma}$$

$$= \frac{1}{4} \frac{e^4}{s^2} \text{Tr}[\not{p}_1 \gamma^\rho \not{p}_2 \gamma^\mu] \text{Tr}[(\not{p}_3 + m_\mu) \gamma^\nu (\not{p}_4 - m_\mu) \gamma^\sigma] \eta_{\mu\nu} \eta_{\rho\sigma}$$

$$\text{Tr}[\not{p}_1 \gamma^\rho \not{p}_2 \gamma^\mu] = p_1^\alpha p_2^\beta \text{Tr}[\gamma_\alpha \gamma^\rho \gamma_\beta \gamma^\mu] =$$

$$= p_1^\alpha p_2^\beta 4(\eta_{\alpha\rho} \eta_{\beta}^\mu - \eta_{\alpha\beta} \eta_{\rho}^\mu + \eta_{\alpha}^\mu \eta_{\rho\beta})$$

$$= 4(p_1^\rho p_2^\mu - \eta^{\rho\mu} (p_1 \cdot p_2) + p_1^\mu p_2^\rho)$$

$$\text{Tr}[(\not{p}_3 + m_\mu) \gamma^\nu (\not{p}_4 - m_\mu) \gamma^\sigma] =$$

Traces with odd # of $\gamma_s = 0$

$$\text{Tr}[\not{p}_3 \gamma^\nu \not{p}_4 \gamma^\sigma - m_\mu \not{p}_3 \gamma^\nu \gamma^\sigma + m_\mu \gamma^\nu \not{p}_4 \gamma^\sigma - m_\mu^2 \gamma^\nu \gamma^\sigma] =$$

$$= p_3^\alpha p_4^\beta \text{Tr}[\gamma_\alpha \gamma^\nu \gamma_\beta \gamma^\sigma] - m_\mu^2 \text{Tr}[\gamma^\nu \gamma^\sigma]$$

$$\text{Tr}[\gamma^\nu \gamma^\sigma] = 4 \eta^{\nu\sigma}$$

$$= 4 p_3^\alpha p_4^\beta (\eta_{\alpha}^{\sigma} \eta_{\beta}^{\nu} - \eta_{\alpha\beta} \eta^{\sigma\nu} + \eta_{\alpha}^{\nu} \eta^{\sigma\beta}) - 4 m^2 \eta^{\nu\sigma}$$

$$= 4 (p_3^{\sigma} p_4^{\nu} - \eta^{\sigma\nu} (p_3 \cdot p_4) + p_3^{\nu} p_4^{\sigma}) - 4 m^2 \eta^{\nu\sigma}$$

$$\langle |M|^2 \rangle = \frac{16}{4} \frac{e^4}{s^2} (p_1^{\rho} p_2^{\mu} - \eta^{\rho\mu} (p_1 \cdot p_2) + p_1^{\mu} p_2^{\rho}) (p_3^{\sigma} p_4^{\nu} - \eta^{\sigma\nu} (p_3 \cdot p_4) + p_3^{\nu} p_4^{\sigma} - m^2 \underbrace{\eta^{\nu\sigma}}_{\eta_{\mu\nu} \eta_{\rho\sigma}})$$

let $\eta_{\mu\nu} \eta_{\rho\sigma}$ act

$$\langle |M|^2 \rangle = \frac{16}{4} \frac{e^4}{s^2} \left[(p_1^{\rho} p_2^{\mu} - \underbrace{\eta^{\rho\mu}}_{\eta_{\rho\mu} = \eta_{\mu\rho}} (p_1 \cdot p_2) + p_1^{\mu} p_2^{\rho}) \times \right. \\ \left. (p_3^{\rho} p_4^{\mu} - \underbrace{\eta^{\sigma\mu}}_{\eta_{\rho\mu} = \eta_{\mu\rho}} \eta_{\mu\nu} \eta_{\rho\sigma} (p_3 \cdot p_4) + p_3^{\mu} p_4^{\rho} - m^2 \underbrace{\eta^{\nu\sigma}}_{\eta_{\mu\nu} \eta_{\rho\sigma}} \eta_{\mu\rho}) \right]$$

$$= \frac{16}{4} \frac{e^4}{s^2} \left[(p_1^{\rho} p_2^{\mu} - \eta_{\mu\rho} (p_1 \cdot p_2) + p_1^{\mu} p_2^{\rho}) (p_3^{\rho} p_4^{\mu} - \eta_{\mu\rho} (p_3 \cdot p_4 + m^2) + p_3^{\mu} p_4^{\rho}) \right] =$$

$$= \frac{16}{4} \frac{e^4}{s^2} \left[(p_1 \cdot p_3)(p_2 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3) - (p_1 \cdot p_2)(p_3 \cdot p_4 + m^2) + \right. \\ \left. + (p_1 \cdot p_4)(p_2 \cdot p_3) + (p_1 \cdot p_3)(p_2 \cdot p_4) - (p_1 \cdot p_2)(p_3 \cdot p_4 + m^2) + \right. \\ \left. - (p_1 \cdot p_2)(p_3 \cdot p_4) - (p_1 \cdot p_2)(p_3 \cdot p_4) + 4(p_1 \cdot p_2)(p_3 \cdot p_4 + m^2) \right] =$$

$$= \frac{8}{4} \frac{e^4}{s^2} \left[(p_1 \cdot p_3)(p_2 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3) - (p_1 \cdot p_2)(p_3 \cdot p_4) + (p_1 \cdot p_2)(p_3 \cdot p_4 + m^2) \right]$$

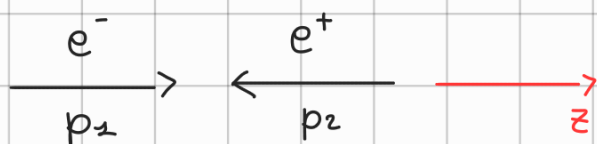
we obtain

$$\rightarrow \langle |M|^2 \rangle = \frac{8e^4}{s^2} \left[(p_1 \cdot p_3)(p_2 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3) + m^2 (p_1 \cdot p_2) \right]$$

we have to express momenta p_i as function of s, θ in the c.o.m. frame

Momenta

- 1) Since $m_e \approx 0 \rightarrow E^2 = p^2 + m^2 \rightarrow E \sim |p|$ (only for electrons / positrons)
- 2) in the c.o.m. frame imagine a scattering on z axis



because of (1)

$$\vec{p}_1 = (0, 0, |p|) = (0, 0, E)$$

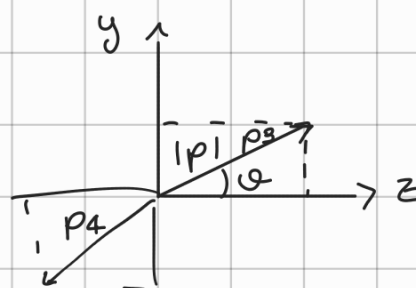
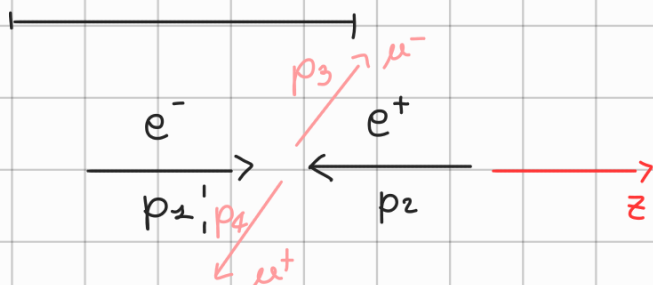
$$\vec{p}_2 = (0, 0, -|p|) = (0, 0, -E)$$

four-vectors have shape $p = (E, p_x, p_y, p_z)$
in this case we have

$$e^-) \quad p_1 = (E, 0, 0, E)$$

$$\rightarrow s = (p_1 + p_2)^2 = (2E, 0, 0, 0)^2 = 4E^2$$

$$e^+) \quad p_2 = (E, 0, 0, -E)$$



$$p \cos \theta = p_z$$

$$p \sin \theta = p_y$$

After the scattering we produce μ^+, μ^- with mom. p_3, p_4
for momentum conservation we must have $p_3 = -p_4$

from trigonometry, assuming

only y, z components, we have:

$$p_3 = (E, 0, p \sin \theta, p \cos \theta)$$

$$p_4 = (E, 0, -p \sin \theta, -p \cos \theta)$$

note: at the leading order $e^+e^- \mu^+\mu^-$

has a cylindrical symmetry, with respect to beam axis (z).

σ does not depend on φ

for muons $m_\mu \neq 0$ so

$$E^2 = p^2 + m^2 \rightarrow p = \sqrt{E^2 - m_\mu^2}$$

← we know the 1!

So:

$$p_1 = (E, 0, 0, E)$$

$$s = 4E^2$$

$$E = \frac{\sqrt{s}}{2}$$

$$p_2 = (E, 0, 0, -E)$$

should we write them?

$$p_3 = (E, 0, \sqrt{E^2 - m_\mu^2} \sin \theta, \sqrt{E^2 - m_\mu^2} \cos \theta) = \left(\frac{\sqrt{s}}{2}, 0, \sqrt{\frac{s}{4} - m^2} \sin \theta, \sqrt{\frac{s}{4} - m^2} \cos \theta \right)$$

$$p_4 = (E, 0, -\sqrt{E^2 - m_\mu^2} \sin \theta, -\sqrt{E^2 - m_\mu^2} \cos \theta)$$

Now we have to express $\langle |M|^2 \rangle$ in terms of θ

$$\langle |M|^2 \rangle = \frac{8e^4}{s^2} \left[(p_1 \cdot p_3)(p_2 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3) + m^2 (p_1 \cdot p_2) \right] =$$

$$(p_1 \cdot p_3) = E^2 - E \sqrt{E^2 - m_\mu^2} \cos \theta$$

$$(p_2 \cdot p_4) = E^2 - E \sqrt{E^2 - m_\mu^2} \cos \theta$$

$$(p_1 \cdot p_4) = E^2 + E \sqrt{E^2 - m_\mu^2} \cos \theta$$

$$(p_2 \cdot p_3) = E^2 + E \sqrt{E^2 - m_\mu^2} \cos \theta$$

$$(p_1 \cdot p_2) = E^2 - (-E^2) = 2E^2$$

same $\left[\begin{array}{l} \text{special relativity} \\ a \cdot b = a_0 b_0 - \vec{a} \cdot \vec{b} \end{array} \right]$

factor 2 in the $\langle |M|^2 \rangle$

same

substituting momenta p_i (function of s, θ) we get:

$$\begin{aligned}
 \langle |M|^2 \rangle &= \frac{8e^4}{s^2} \left[2(E^2 + E\sqrt{E^2 - m_\mu^2} \cos\theta)^2 + 2(E^2 - E\sqrt{E^2 - m_\mu^2} \cos\theta)^2 + 2m^2 E^2 \right] \\
 &= \frac{2 \cdot 8e^4}{s^2} \left[E^4 + 2E^3 \sqrt{E^2 - m_\mu^2} \cos\theta + E^2(E^2 - m_\mu^2) \cos^2\theta + \right. \\
 &\quad \left. E^4 - 2E^3 \sqrt{E^2 - m_\mu^2} \cos\theta + E^2(E^2 - m_\mu^2) \cos^2\theta + m^2 E^2 \right] \\
 &= \frac{16e^4}{s^2} \left[E^4 + E^2 \underbrace{(E^2 - m_\mu^2)}_{k^2} \cos^2\theta + m^2 E^2 \right] \\
 &= \frac{16e^4}{s^2} E^2 \left[(E^2 + m_\mu^2) + (E^2 - m_\mu^2) \cos^2\theta \right] \quad E^2 = \frac{s}{4} \\
 &= \frac{16e^4}{s^2} \frac{s}{4} \frac{s}{4} \left[\left(1 + \frac{4m_\mu^2}{s}\right) + \left(1 - \frac{4m_\mu^2}{s}\right) \cos^2\theta \right] \\
 &= e^4 \left[\left(1 + \frac{4m_\mu^2}{s}\right) + \left(1 - \frac{4m_\mu^2}{s}\right) \cos^2\theta \right]
 \end{aligned}$$

differential
the cross section

now that we have $\langle |M|^2 \rangle$ we can write $\frac{d\sigma}{d\Omega}$

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s} \frac{|P_f|}{|P_i|} \langle |M|^2 \rangle$$

$|P_i| = E$ $m_e \approx 0$
not sure about initial & final momenta

$$|P_f| = \sqrt{E^2 - m_\mu^2} \quad m_\mu \neq 0$$

$$\frac{P}{E} = \beta = \sqrt{1 - \frac{4m^2}{s}}$$

$$= \frac{1}{64\pi^2 s} \frac{\frac{\sqrt{s}}{2} \sqrt{1 - \frac{4m^2}{s}}}{\frac{\sqrt{s}}{2}} e^4 \left[\left(1 + \frac{4m_\mu^2}{s}\right) + \left(1 - \frac{4m_\mu^2}{s}\right) \cos^2\theta \right]$$

substituting the fine structure constant α :

$$\begin{aligned}
 \frac{d\sigma}{d\Omega} &= \frac{e^4}{64\pi^2 s} \sqrt{1 - \frac{4m^2}{s}} \left[\left(1 + \frac{4m_\mu^2}{s}\right) + \left(1 - \frac{4m_\mu^2}{s}\right) \cos^2\theta \right] \\
 &= \underbrace{\frac{\alpha^2}{4s}}_{\beta} \sqrt{1 - \frac{4m^2}{s}} \left[\left(1 + \frac{4m_\mu^2}{s}\right) + \left(1 - \frac{4m_\mu^2}{s}\right) \cos^2\theta \right]
 \end{aligned}$$

$\beta \sim 0.75$ imposing $\sqrt{s} = 3m_\mu$ not ultrarelativistic, as expected!

lab frame

3)

e^- at rest $p_1 = (0, 0, 0, 0)$

$$s = (p_1 + p_2)^2 = p_1^2 + p_2^2 + 2 p_1 \cdot p_2 = 2m_e^2 + 2 \left(\overset{m_e}{\downarrow} E_2 E_2 - \vec{p}_1 \cdot \vec{p}_2 \right)$$

if e^- at rest $\rightarrow \vec{p}_1 = 0$

$$\rightarrow s = 2m_e^2 + 2m_e E_2 \rightarrow E_2 = \frac{s - 2m_e^2}{2m_e}$$

$$E_2^2 = p_2^2 + m^2$$
$$p_2 = \sqrt{E^2 - m^2}$$

$$\beta = \frac{p_{\text{TOT}}}{E_{\text{TOT}}} = \frac{p_2}{E_2 + m_e} = \frac{\sqrt{E_2^2 - m_e^2}}{E_2 + m_e}$$

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}}$$

$$p_{x-\text{lab}} = p_{x-\text{cm}}$$

$$p_{y-\text{lab}} = p_{y-\text{cm}}$$

boost only at z !

$$p_{z-\text{lab}} = \gamma(p_{z-\text{cm}} + \beta E_{-\text{cm}})$$

$$E_{-\text{lab}} = \gamma(E_{-\text{cm}} + \beta p_{z-\text{cm}})$$